RDFIA 3 - b

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1 Introduction

This lab builds on the previous one with a major change: the posterior distribution is no longer linked to the Gaussian prior. As a result, new techniques need to be used, which are explored in this lab. These include the Maximum A Posteriori (MAP) estimate, the Laplace approximation and the variatonial inference with mean-field approximation and Monte Carlo Dropout.

2 Part 1 - Bayesian Logistic Regression

2.1 Maximum-A-Posteriori Estimate

Question 1-1

We have the equality for the predictive distribution :

$$p(y = 1|x, D) = \int p(y = 1|x, w)p(w|D)dw$$

And if we approximate $p(w|D)$ with a Dirac delta function $\delta(w - w_{MAP})$, we get that :

$$p(y = 1|x, D) \approx p(y = 1|x, w_{MAP})$$

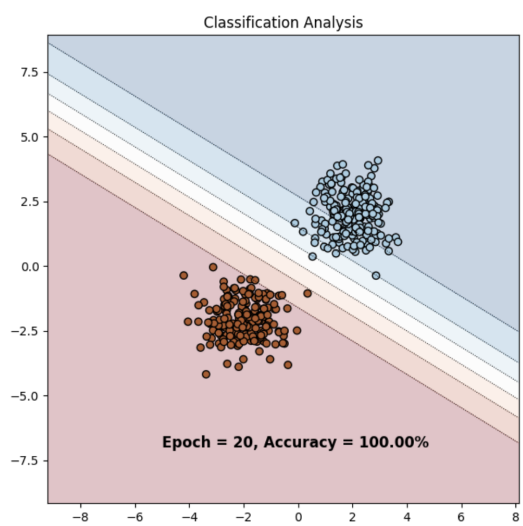


Figure 1: Classification analysis for Maximum A-Posteriori approximation

As we can see on the FIGURE 1, the decision boundary remains linear, and the model's uncertainty does not grow as we move away from the training data. Essentially, the model provides a confidence measure for the linear boundary but struggles to quantify uncertainty for distant points. This limitation arises because the parameter estimates can confidently classify points within the training distribution but fail to capture meaningful uncertainty for underrepresented data regions.

2.2 Laplace Approximation

Question 1-2

Bayesian Logistic Regression with the Laplace approximation provides a better representation of uncertainty in model parameters. As shown in the FIGURE 2, in regions near the two clusters, the model confidently classifies points. However, in areas with sparse or no data, the model's uncertainty increases, highlighting epistemic uncertainty related to the model's limited knowledge of unseen regions. This behavior seems more adapted compared to the linear boundary obtained in the previous MAP estimate.

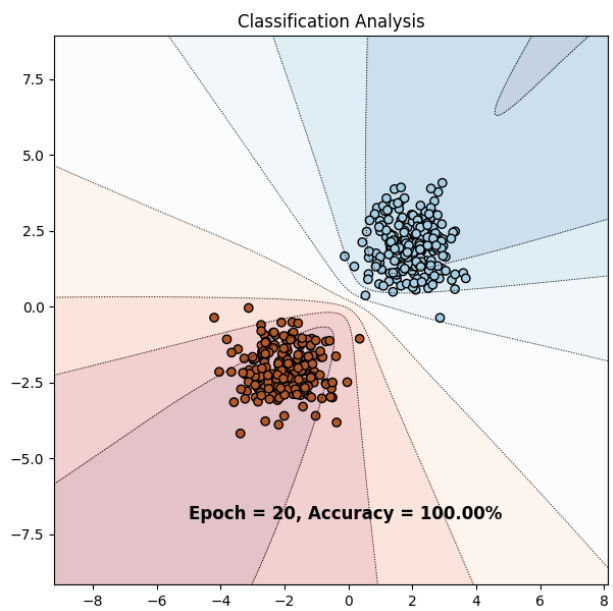
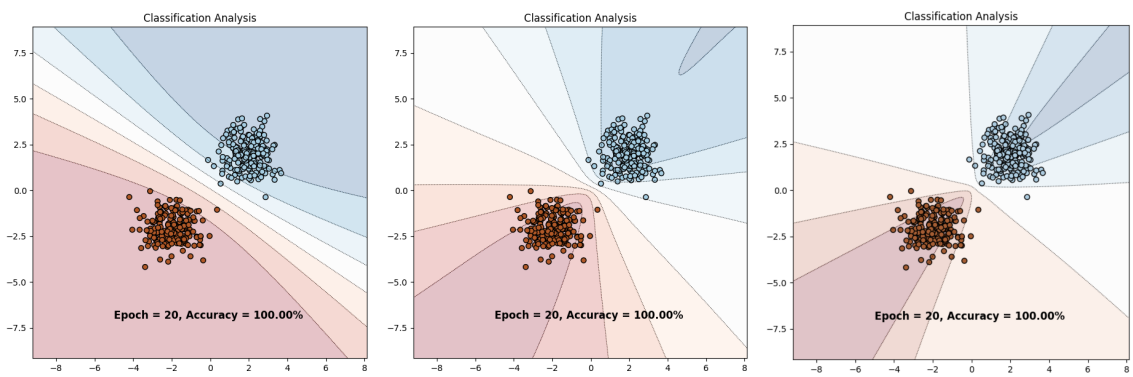


Figure 2: Classification analysis for Laplace Approximation

Question 1-3

The FIGURE 3 presents the influence of the *Weight Decay* parameter on the regression. As we can see, the uncertainty far from the training distribution increases when the *Weight Decay* decreases. This can be explained by the fact that when the *Weight Decay* decreases, the prior over the model weights becomes less restrictive, allowing the weights to take larger values based on the training data. This increased flexibility enables the model to fit the training data more closely, but it also leads to higher epistemic uncertainty in regions far from the training data. In these areas, where the model has little to no information, the lack of strong prior regularization means the model relies heavily on the learned weights, which may extrapolate unpredictably.



(a) Weight Decay $= 5$ (b) Weight Decay $= 5 * 10^{-2}$ (c) Weight Decay $= 5 * 10^{-4}$

Figure 3: Analysis of Weight Decay's influence

2.3 Variational Inference

Question 1-4

The LinearVariational class represents a single linear layer with variational inference, where the weights and biases are modeled as probability distributions rather than fixed values. It is initialized with learnable parameters for the mean and a transformed variance of the weights (w_μ, w_ρ) and biases (b_μ), along with a prior variance $(prio_{std})$ reflecting prior beliefs about the weight distribution. The sampling method uses the reparameterization trick to draw samples from the variational posterior, enabling gradient-based optimization despite stochastic sampling. The kl divergence

method computes the Kullback-Leibler divergence between the variational posterior and the prior, quantifying the difference between the learned distribution and the prior belief. The forward method performs a linear transformation using sampled weights and the mean bias.

The VariationalLogisticRegression class builds on LinearVariational to implement logistic regression with variational inference. It consists of a single variational linear layer for the transformation, and its forward method applies the sigmoid function to the linear output for classification. The KL divergence term is obtained by invoking the corresponding method from the LinearVariational layer to regularize the loss function.

Question 1-5

The training loop follows a standard approach, where the objective is to maximize the Evidence Lower Bound (ELBO). Directly maximizing the data likelihood is often impractical due to the complexity of integrating over the weight distribution. Instead, the loss function includes the Kullback-Leibler (KL) divergence, which measures the difference between the variational posterior $q_\theta(w)$ and the prior $p(w)$, acting as a regularization term that encourages the learned distribution to remain close to the prior.

To ensure a good fit to the data, the model computes the negative log-likelihood using a binary cross-entropy loss. Since maximizing ELBO is equivalent to minimizing the sum of NLL and KL divergence, gradient descent is used to optimize the variational parameters, bringing the approximation closer to the true posterior.

The variational approach results in decision boundaries that balance between the Maximum a Posteriori (MAP) estimate and the Laplace approximation. The model effectively handles noisy data points, ensuring that all training points fall within high-confidence regions. As shown in the Figure 4, the decision boundary adapts to data uncertainty, offering a distinctive frontier in the central area while capturing aleatoric uncertainty effectively.

The variational approximation tries to closely match the true posterior by adjusting a flexible distribution and minimizing the KL divergence. This process involves solving a more complex optimization problem. In contrast, the Laplace approximation takes a simpler approach by fitting a Gaussian distribution around the MAP estimate, using the shape of the log-posterior at that point. It assumes the posterior is roughly Gaussian near the MAP estimate and mainly focuses on finding this estimate and computing the curvature.

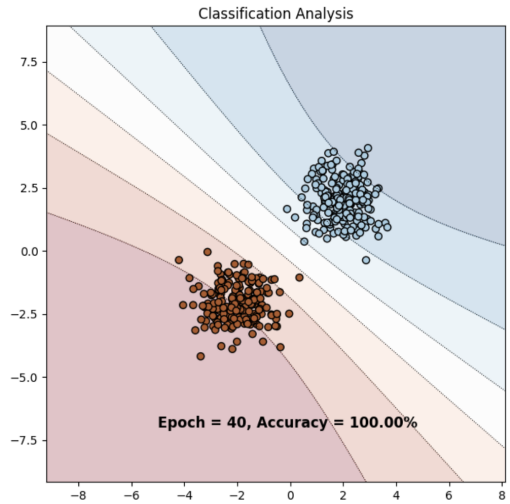


Figure 4: Classification analysis for Variational inference

3 Part 2 - Bayesian Neural Networks

3.1 Variational Inference with Neural Networks

This part studies the effect of using a Neural Network with variational inference. The test is performed on a more complex dataset shaped like semi-circles.

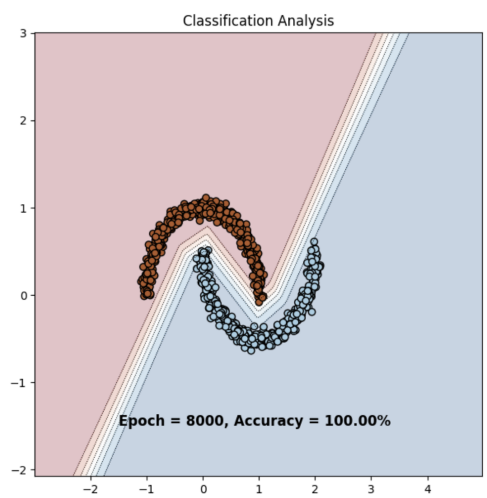


Figure 5: Classification analysis for Variational Inference with Bayesian Neural networks

The FIGURE 5 shows very satisfying results: the Neural Network is able to capture the complex characteristics of the data and to classify the points with a high level of confidence. We also notice thanks to FIGURE 6 that the model converges pretty quickly and not many epochs are needed to classify the dataset efficiently.

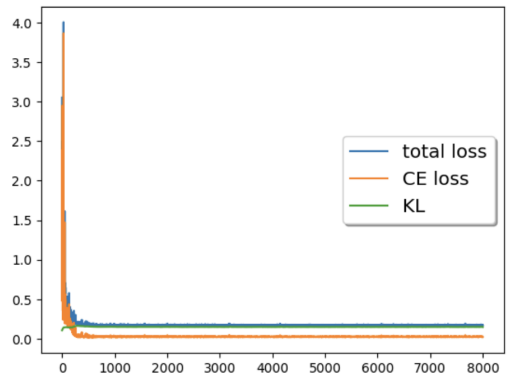


Figure 6: Loss function (Hidden size = 50)

Moreover, the FIGURE 8 illustrates the role of the hidden size in the neural network: a higher hidden size enables the model to capture more specific patterns in the data and to classify it efficiently with higher confidence. Each neuron's weight added to the Network is treated as a random variable and contributes to determine the distribution of the data with more precision. However, increasing the hidden size makes the model more complex and requires more time to train but also more epochs as we see in the FIGURE 7 where the model needs way more time to converge than on the FIGURE 6].

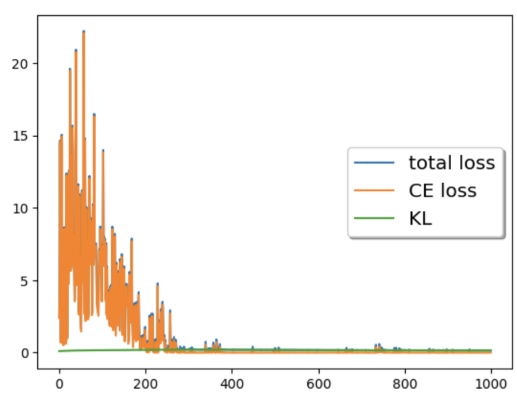
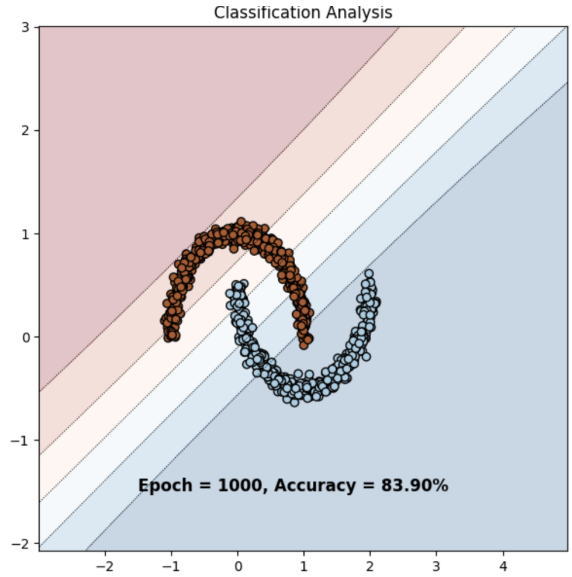
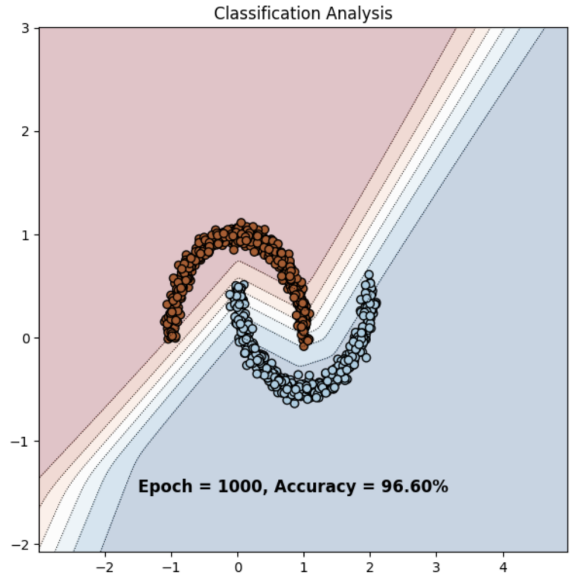


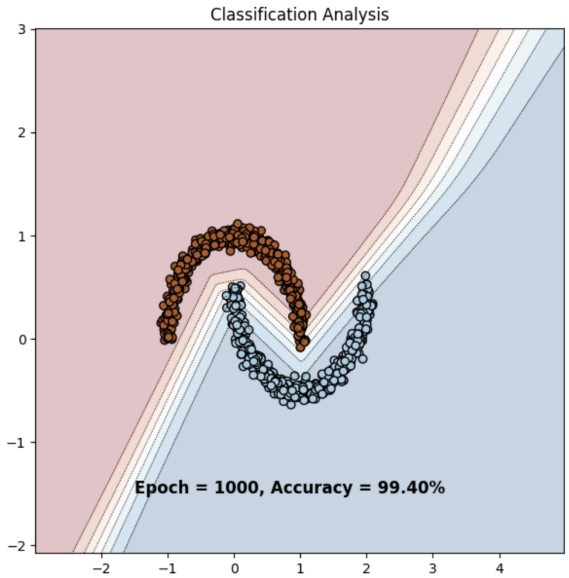
Figure 7: Loss function (Hidden size = 500)



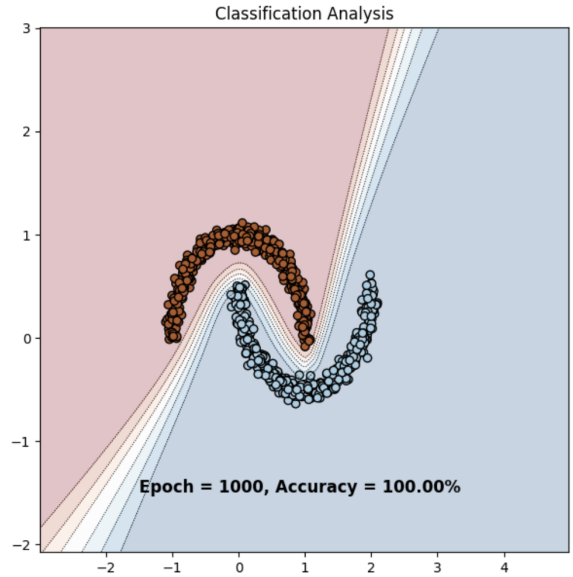
(a) Hidden size = 1



(b) Hidden size = 10



(c) Hidden size = 20



(d) Hidden size = 200

Figure 8: Analysis of hidden size's influence

3.2 Monte Carlo Dropout

The results of the FIGURE 9 are obtained when setting the dropout layer to *training* = *True* with $p = 0.9$ as asked in the practical.

The results show decision boundaries with surrounding uncertainty bands created by applying dropout at inference and averaging predictions across multiple passes. These bands highlight the model's confidence levels, with areas of higher uncertainty appearing more dispersed. Unlike deterministic models that produce smooth decision boundaries, MC Dropout introduces variability that reflects uncertainty across different input regions.

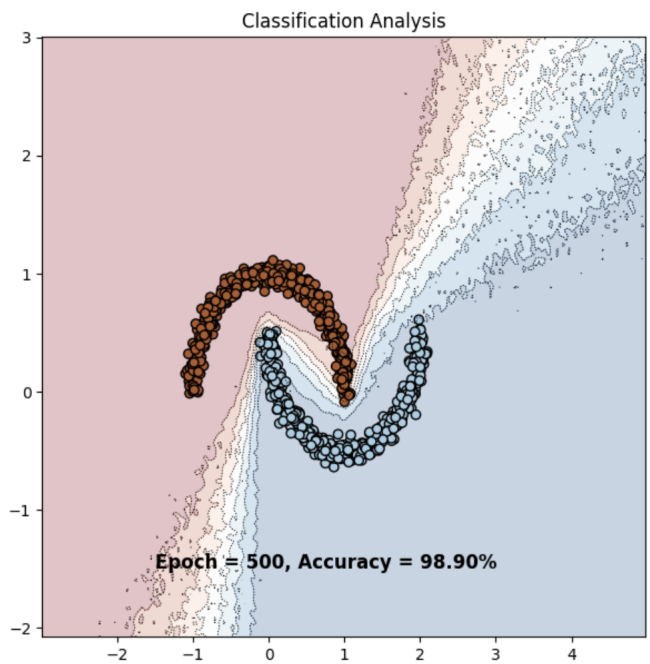


Figure 9: Classification analysis with Monte Carlo dropout

MC Dropout offers a more efficient and flexible approach compared to Bayesian Logistic Regression with variational inference. By using the dropout mechanism during both training and inference, it approximates the posterior distribution without the need to optimize complex variational distributions, significantly reducing computational overhead. This method can be easily integrated into standard neural networks by simply adding dropout during inference, allowing for uncertainty estimation in predictions. In contrast, Bayesian Logistic Regression with variational inference requires defining and optimizing a variational distribution to approximate

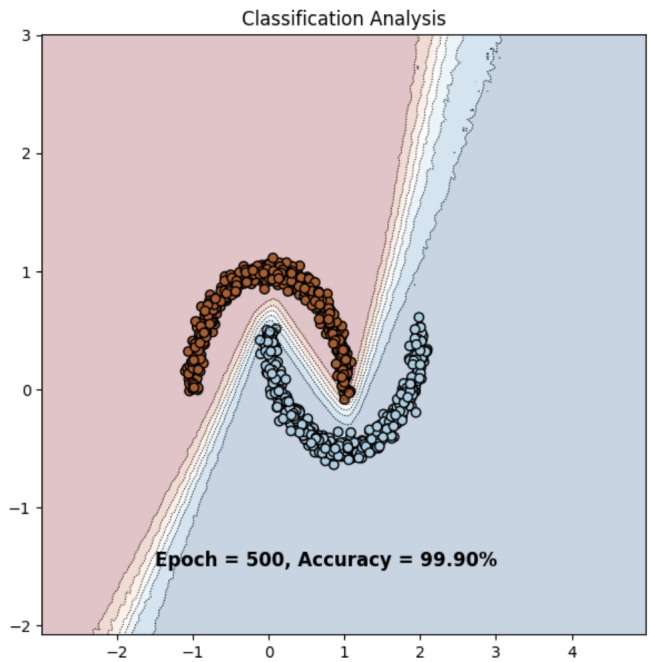


Figure 10: Results obtained by MC Dropout

the posterior, which is limited to simpler models like logistic regression.